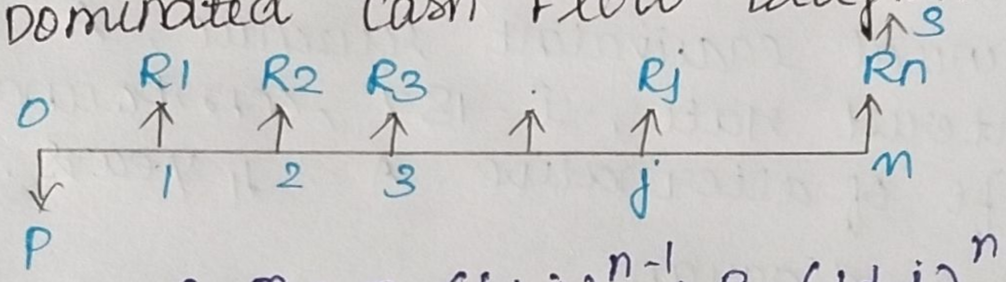
1) EXPLAIN FUTURE WORTH METHOD OF COMPARISON WITH SUITABLE EXAMPLE :

* In the Future worth method of comparison of alternatives, the future worth of various alternatives will be computed

* Then, the alternative, with the maximum future worth of net revenue or with maximum future worth of net cost will be selected as best alternative for implementation.

Revenue Dominated Cash Flow Diagram.

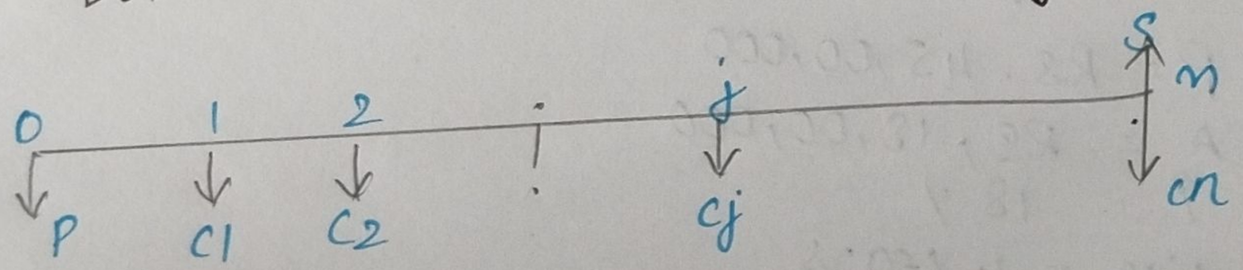


Formula : $-P(1+i)^n + R_1(1+i)^{n-1} + R_2(1+i)^{n-2} + \ldots$
$Rj(1+i)^{n-j} + \ldots Rn + S$

* In the Above formula, the expenditure is assigned with negative sign and the revenues are assigned with positive sign.

* P represents an initiate investment, Rj is the net-revenue at end of jth year and S is salvage value at end of n th year.

Cost Dominated cash Flow Diagram.



Formula : $FW(i) = P(1+i)^n + C_1(1+i)^{n-1} + C_2(1+i)^{n-2} + \ldots$
$+ Cj(1+i)^{n-j} + Cn - S$

* In this formula, the expenditures are assigned with postive sign & revenues with negative sign.

Example :

consider the following 2 mutually exclusive alternatives

ALTERNATIVE	END OF YEAR				
	0	1	2	3	4
A (Rs.)	-5000000	2000000	2000000	2000000	2000000
B (Rs.)	-4500000	1800000	1800000	1800000	1800000

At $i = 18\%$, Select the best alternative based on Future worth method of comparison.

SOLN : Alternative A :

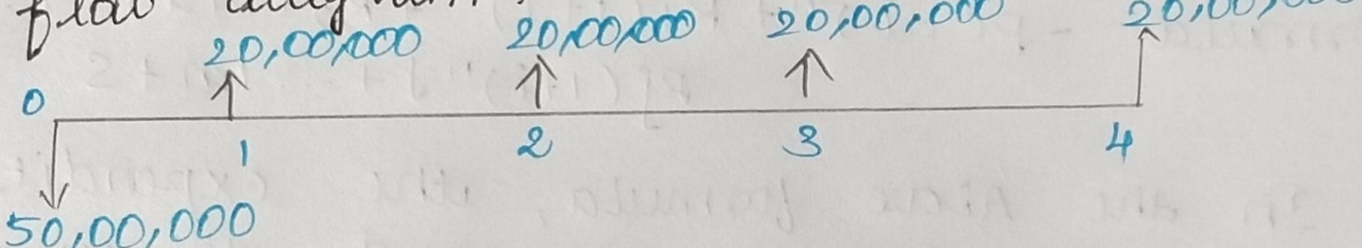
Initial investment , $P = 50,00,000$

Annual equivalent revenue , $A =$ Rs. 20,00,000

Interest rate , $i = 18\%$, compounded annually

Life of alternative A = 4 years.

cash flow diagram :-



Future worth amount of A :

$$FW_A(18\%) = -50,00,000\ (F/P,\ 18\%,4) + 20,00,000\ (F/A,\ 18\%,4)$$

$$= -50,00,000\ (1.939) + 20,00,000\ (5.215)$$

$$= \text{Rs. } 7,35,000$$

Alternative B :

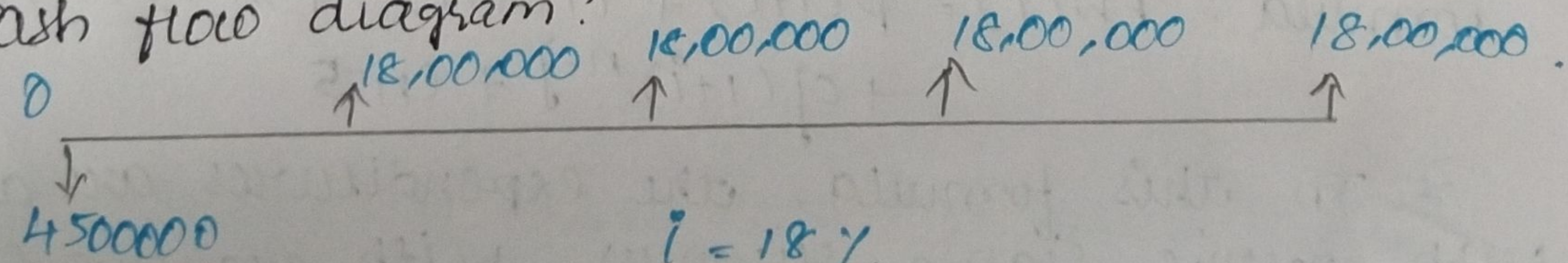
$P =$ Rs. 45,00,000

$A =$ Rs, 18,00,000

$i = 18\%$

Life = 4 years

cash flow diagram :



Future worth of B:

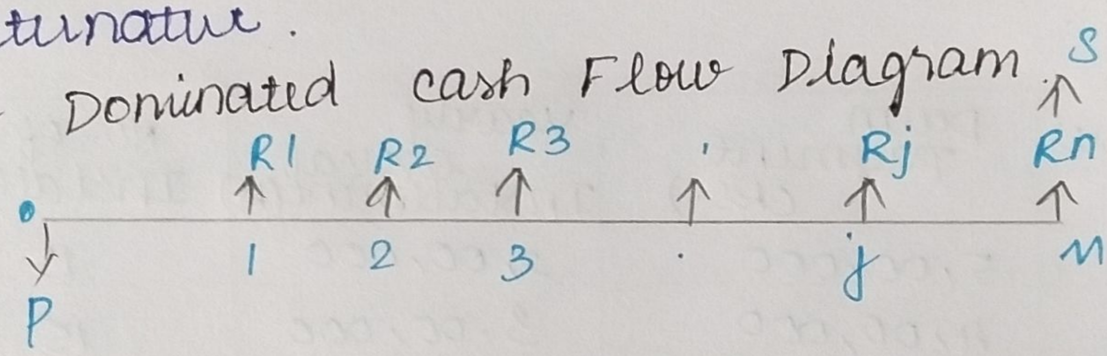
$$FW_B(18\%) = -4500{,}000\,(F/P, 18\%, 4) + 18{,}00{,}000\,(F/A, 18\%, 4)$$
$$= -4500{,}000\,(1.939) + 18{,}00{,}000\,(5.215)$$
$$= Rs.\ 6{,}61{,}500$$

* The Future worth of alternatives A is greater than B. Thus, Alternatives A should be selected.

2) EXPLAIN ANNUAL EQUIVALENT METHOD OF COMPARISON WITH SUITABLE EXAMPLE:

* In Annual equivalent method of comparison, first the annual equivalent cost or the revenue of cash each alternative will be computed.
* Then, the alternative with the maximum annual equivalent revenue in the case of revenue based comparison will be selected as best alternative.

Revenue Dominated cash Flow Diagram



$P \rightarrow$ Initial investment
$R_j \rightarrow$ net revenue at end of jth year
$S \rightarrow$ Salvage value at end of n^{th} year

Formula:-

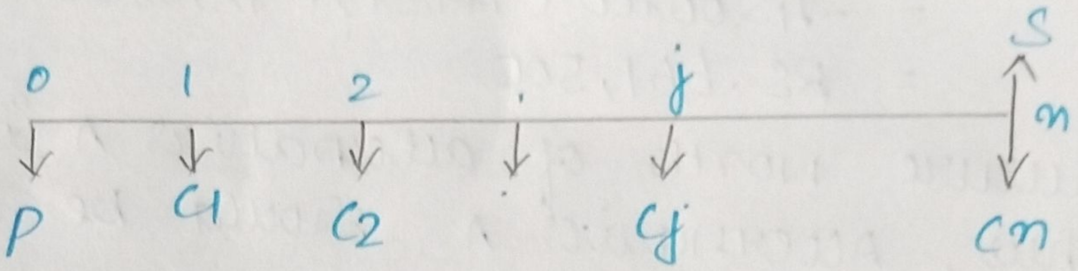
$$PW(i) = -P + R_1/(1+i)^1 + R_2/(1+i)^2 + \ldots R_j/(1+i)^j + \ldots + R_n/(1+i)^n + S/(1+i)^n$$

$$A = PW(i)\,\frac{i(1+i)^n}{(1+i)^n - 1}$$
$$= PW(i)(A/P, i, n)$$

where $(A/P, i, n)$ is called equal payment series

Capital recovery factor.

cost Dominated cash flow Diagram.



$$PW(i) = P + C1/(1+i)^1 + C2/(1+i)^2 + \ldots Cj/(1+i)^j + \ldots$$
$$+ Cn/(1+i)^n - S/(1+i)^n$$
$$A = PW(i) \cdot \frac{i(1+i)^n}{(1+i)^n - 1}$$

<u>Example</u>

A company is planning to purchase an advanced machine centre. There original manufacturer have responded to its tender whose particulars are tabulated.

Manufactures	Down Payment (Rs.)	yearly equal Instalment (Rs)	No. of Instalments
1	5,00,000	2,00,000	15
2	4,00,000	3,00,000	15
3	6,00,000	1,50,000	15

Determine the best alternative based on annual equivalent method by assuming $i = 20\%$ compounded annually.

SOLn :* <u>alternative A1</u>:

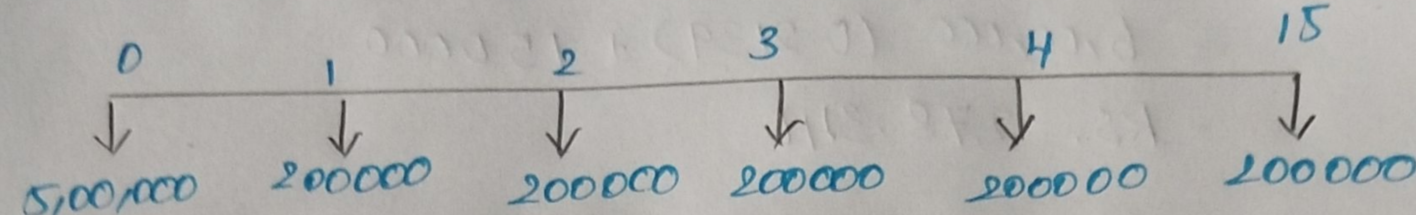
Down payment, P = Rs. 5,00,000

yearly equal installment, A = Rs, 2,00,000

n = 15 years

i = 20%, compounded Annually.

Cash Flow Diagram :



$AE_1 (20\%) = 5,00,000\ (A/P, 20\%, 15) + 2,00,000$

$= 5,00,000\ (0.2139) + 2,00,000$

$= 3,06,950.$

* Alternative 2

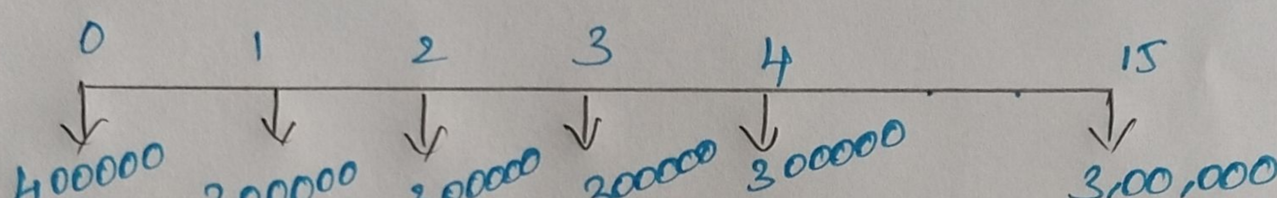
$P = RS.\ 4,00,000$

$A = RS.\ 3,00,000$

$n = 15$ years

$i = 20\%.$

Cash Flow Diagram



$AE_2 (20\%) = 4,00,000\ (A/P, 20\%, 15) + 3,00,000$

$= 4,00,000\ (0.2139) + 3,00,000$

$= RS.\ 3,85,560$

* Alternative 3 :

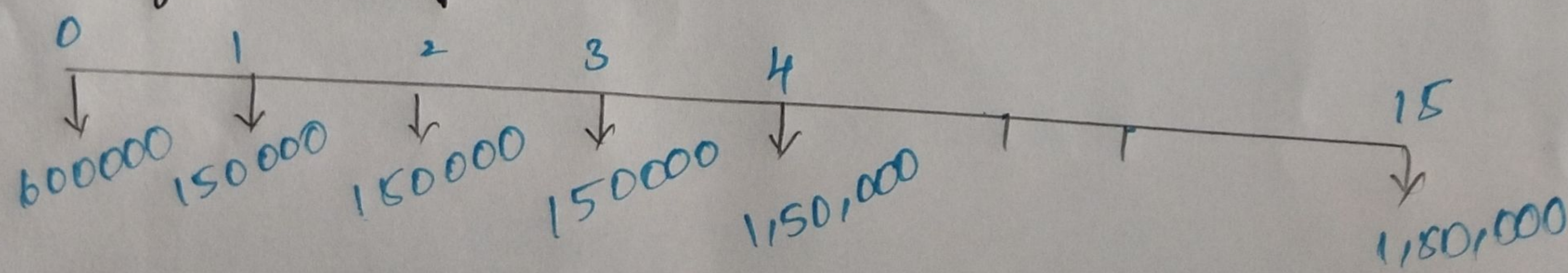
$P = RS.\ 6,00,000$

$A = RS.\ 1,50,000$

$n = 15$ years

$i = 20\%.$

Cash flow Diagram :



$$AE_3(20\%) = 6{,}00{,}000\ (A/P, 20\%, 15) + 1{,}50{,}000$$
$$= 6{,}00{,}000\ (0.2139) + 150000$$
$$= Rs.\ 2{,}78{,}340$$

* The Annual equivalent cost of manufacturer 3 is less than that of 1 and 2.

* Therefore, the company should buy the advanced machine centre from manufacturer 3.